

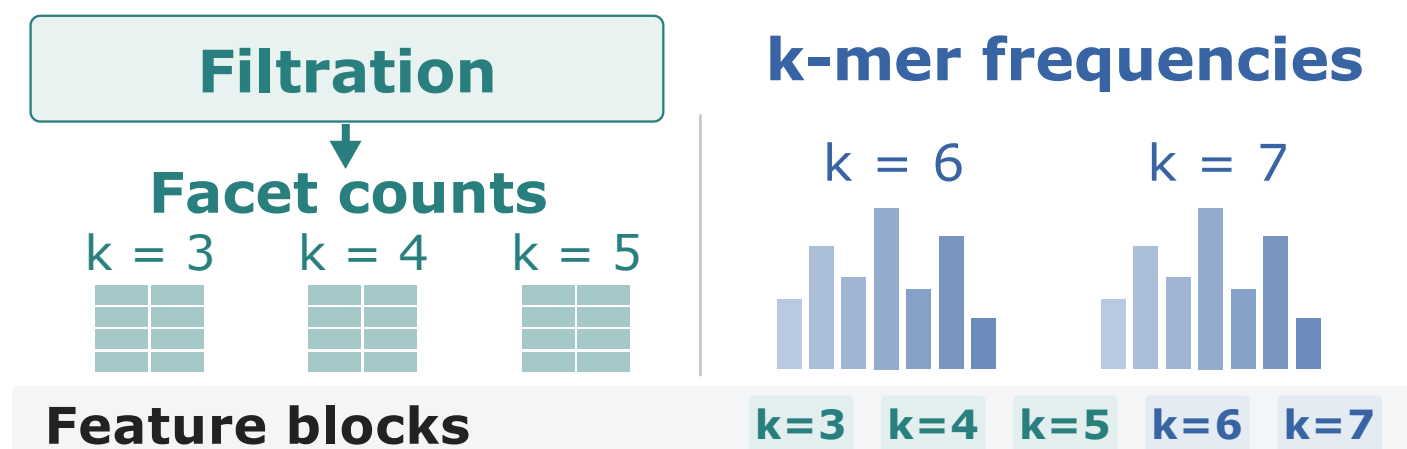
a. Genome sequences

AGGAACTCTGCTACTAGAAT ...
AGTCCTATCCTACGCTTAC ...
CTGGGGTAACGCTAAAATC ...
Family-labeled genomes

b. k-mer positions

Repeated 3-mer: CTA
11 14
AGGAACTCTGCTACTAGAAT
CTA: [11, 14, ...]
Repeat for each k-mer
Starting positions use
1-based indexing

c. Multiscale features



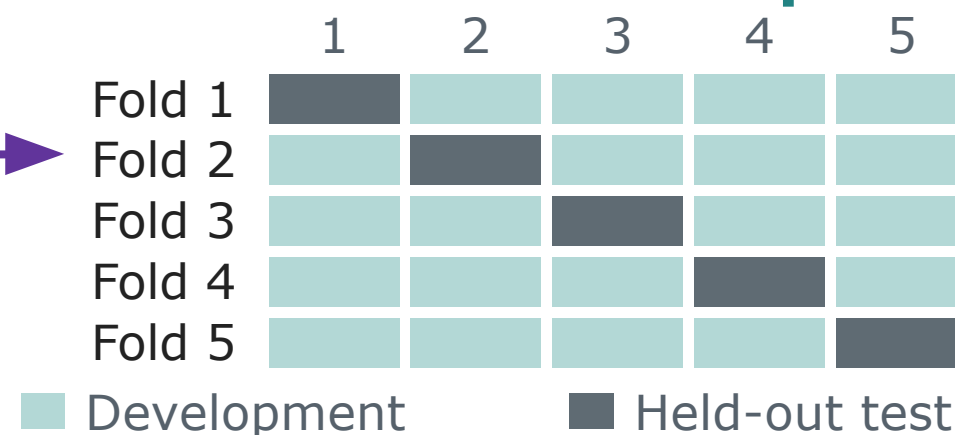
d. Repeated stratified 5-fold cross-validation

Next seed

30 random seeds
1, 2, ..., 30

New stratified partition
for each seed

Stratified 5-fold split



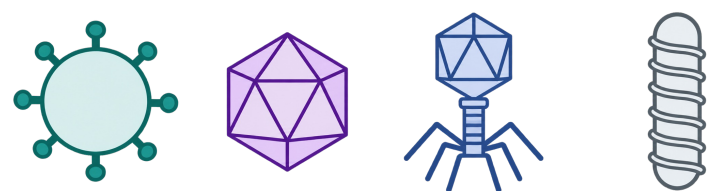
One held-out group per fold

Other four groups provide
training and validation genomes
Facet scaling fitted on training data

Fold-specific normalization

g. Viral classification

KNN (1 or 5 neighbors)



Held-out predictions

After 5 folds

Pool held-out predictions

Compute one result per seed

After all 30 seeds

Mean $\pm$ sample standard deviation

across 30 seed-level results

f. Cosine distances

Training + validation reference

CNN distances

Transformer distances

Mean

Consensus distances

e. Model training and embeddings

All five blocks enter each encoder

CNN encoder

CNN embedding

Transformer encoder

Transformer embedding

One embedding vector per genome, per model

Repeat for all 5 held-out folds